Crop damage after 27 Sep 2025: Madha taluka, Solapur

Pik-drishti AI · satellite screening for panchanama field verification

Item	Value
Villages	119
Cropland	124,161 ha
Damaged cropland	12,579 ha (10.1%)
Villages to verify first	1
Radar passes after the event	2025-09-30, 2025-10-05
Optical images after the event	2025-10-05, 2025-10-07, 2025-10-10, 2025-10-12, 2025-10-14, 2025-10-15, 2025-10-17, 2025-10-20

Method

Flood: Sentinel-1 VV after the event vs the same weeks in 2019-2024 at each pixel; flooded if -2.0 standard deviations below normal and darker than -15.0 dB.

Loss: Sentinel-2 NDVI 7-35 days after the event vs the same weeks in 2019-2024; loss >= 50% severe, 25-50% moderate, counted only where the drop is also unusual for that pixel (-1.5 standard deviations).

Damaged: Damaged crop area = flooded, or moderate/severe NDVI loss.

Flag: Villages with 33% or more of their crop area damaged, or 10% or more flooded, are listed first for field verification (SDRF input subsidy needs 33% crop loss).

Limits: Radar misses water under tall, dense crops; NDVI loss can also come from early harvest. A screening for field verification, not a loss certificate.

Results

Village	Crop area (ha)	Flooded %	Damaged %	Damaged (ha)	Severe %	NDVI loss %	Verify first
Wakav	617.5	19.5	26.5	163.8	22.6	17.6	True
Mhaisgaon	2381.3	2.8	16.8	400.5	9.6	13.9	False
Tembhurni	3777.8	0.2	10.1	381.0	4.8	9.8	False
Laul	3484.7	1.0	8.9	308.6	4.5	9.6	False
Bembale	2349.0	0.9	12.3	288.4	5.0	11.5	False
Manegaon	3399.1	2.1	8.3	280.8	5.2	9.6	False
Kurdu	3342.5	1.5	7.8	261.0	4.2	8.5	False
Chobhepimpari	2051.8	1.1	11.5	235.4	6.3	11.3	False
Upalai Bk.	3259.9	1.2	7.2	234.2	3.8	8.5	False
Ropale Kavhe	2623.1	2.4	8.5	223.8	5.5	10.7	False
Tulshi	2339.9	0.9	9.3	217.5	4.8	11.1	False
Pimpalner	2102.9	0.5	9.7	204.4	5.1	9.5	False

Village	Crop area (ha)	Flooded %	Damaged %	Damaged (ha)	Severe %	NDVI loss %	Verify first
Chandaj	1004.8	0.3	20.1	201.7	8.9	15.8	False
Dhavalas	1438.9	1.1	14.0	201.0	8.3	15.0	False
Lahu	972.6	2.6	19.9	193.7	11.6	15.3	False
Darfal	1468.5	3.8	13.0	191.6	8.9	11.7	False
Tandulwadi	1077.6	6.1	17.6	189.6	11.4	14.5	False
Upalai Kh.	2553.9	1.7	7.2	184.7	4.4	9.1	False
Khairao	1255.8	7.9	14.7	184.4	11.5	11.3	False
Ghoti	1918.7	1.4	9.5	181.7	4.6	9.7	False
Wadshinge	2181.6	2.1	8.3	180.1	5.0	8.7	False
Anjangaon Umate	2113.0	0.9	8.1	171.0	5.0	10.3	False
Kavhe	1950.6	0.7	8.7	170.6	4.7	9.7	False
Dahiwali	1402.7	0.4	12.1	170.0	5.7	10.7	False
Nimgaon [Tembhurni]	1169.9	0.9	14.4	168.1	8.6	12.6	False
Barloni	1920.0	1.3	8.7	167.3	4.5	9.0	False
Kewad	1300.2	5.1	12.9	167.2	8.9	11.7	False
Takali Tembhurni	979.0	0.2	16.2	158.4	7.3	13.9	False
Pimpalkhunte	1285.1	2.1	12.1	155.4	6.8	11.9	False
Kanhergaon	1212.4	0.5	12.1	146.5	5.8	11.7	False
Kumbhej	1004.2	8.3	14.5	146.0	11.3	11.8	False
Anjangaon Kh.	1726.5	1.2	8.4	145.3	4.8	9.4	False
Madha	2131.9	1.6	6.8	144.1	4.0	8.1	False
Upalawate	1352.3	1.4	10.6	143.4	5.8	9.9	False
Sapatne Tembhurni	1387.1	0.6	10.1	139.9	5.4	10.2	False
Undargaon	1395.6	6.0	10.0	139.2	7.9	9.1	False
Alegaon Bk	794.3	0.3	17.3	137.3	8.3	14.2	False
Aran	2341.4	0.8	5.7	134.1	2.8	6.8	False
Shiral Tembhurni	814.9	0.2	15.7	127.6	9.1	15.1	False
Bhend	1455.2	1.6	8.6	124.8	4.9	9.2	False
Papnas	518.9	8.4	24.1	124.8	15.7	18.1	False
Ujani Madha	1173.1	0.8	10.5	123.5	5.5	11.3	False
Alegaon Kh	628.0	0.2	19.2	120.3	8.7	15.6	False
Adhegaon	799.9	0.1	14.9	119.4	7.0	12.6	False

Village	Crop area (ha)	Flooded %	Damaged %	Damaged (ha)	Severe %	NDVI loss %	Verify first
Nimgaon (Madha)	763.2	6.2	15.5	118.1	10.4	12.9	False
Ridhore	847.3	2.6	13.8	117.0	7.3	11.1	False
Akole Kh	776.4	0.3	15.1	116.9	6.3	13.1	False
Hole Kh.	1142.7	0.7	10.1	115.2	5.2	10.4	False
Parite	1137.4	1.8	9.8	111.7	4.1	9.7	False
Akole Bk	1109.9	1.4	9.9	109.6	4.6	8.9	False
Shevare	808.5	0.2	12.2	99.0	5.8	12.7	False
Ranzani	595.6	0.2	16.5	98.4	8.8	15.2	False
Nagorli	769.8	0.2	11.8	91.1	5.6	10.6	False
Sapatne (Bhose)	1249.1	1.2	7.3	90.7	3.9	8.5	False
Wadachiwadi	1124.6	1.2	8.0	90.4	4.7	10.1	False
Akulgaon	1106.8	1.5	7.9	87.7	4.4	9.2	False
Sultanpur	431.1	6.8	20.3	87.5	14.9	15.6	False
Warawade	1099.9	1.2	7.9	86.5	3.7	8.8	False
Gar Akole	553.4	0.5	15.5	85.7	6.1	12.8	False
Padasali	1072.4	1.4	7.9	84.8	4.2	9.7	False
Bhosare	1098.2	1.8	7.7	84.7	4.6	8.9	False
Bitergaon	863.4	1.4	9.6	83.3	6.2	10.8	False
Bavi	1184.9	0.6	6.9	81.8	3.6	9.2	False
Malegaon	744.3	0.1	10.9	81.0	6.2	11.7	False
Loni	688.0	1.2	11.5	79.0	7.1	12.6	False
Chinchgaon	910.1	1.0	8.7	78.7	5.0	9.1	False
Wadoli	487.0	0.1	15.8	77.2	7.1	13.8	False
Ambad	622.4	1.2	12.4	77.1	7.4	13.8	False
Shiral Madha	985.0	1.8	7.8	77.1	4.4	8.1	False
Bhutashte	1262.8	0.9	6.0	76.1	3.3	7.9	False
Shedshinge	791.7	1.0	9.3	73.5	5.1	9.8	False
Ahergaon	753.8	0.9	9.6	72.4	4.3	9.9	False
Tadavale	889.9	1.9	8.1	72.1	5.1	9.6	False
Paritewadi	1035.2	1.4	6.9	71.3	3.9	7.4	False
Tambave	613.6	0.7	11.6	71.1	5.5	10.3	False
Modnimb	1167.6	0.6	5.7	66.0	3.2	6.7	False

Village	Crop area (ha)	Flooded %	Damaged %	Damaged (ha)	Severe %	NDVI loss %	Verify first
Venegaon	604.1	0.8	10.5	63.7	5.3	9.9	False
Mungashi	600.5	2.0	10.6	63.4	6.3	10.6	False
Chincholi	991.0	1.0	6.2	61.4	3.3	7.5	False
Akumbhe	670.5	1.0	9.1	60.9	3.9	9.5	False

Data sources

Dataset	Provider	Used for	Licence / citation
Sentinel-1 SAR GRD (C-band, VV)	European Space Agency / Copernicus	Flooded and waterlogged cropland after the event (Z-score against 2019-2024)	Copernicus open licence. DeVries et al. (2020), Remote Sensing of Environment 240:111664; UN-SPIDER recommended practice
Sentinel-2 L2A surface reflectance (harmonized) with Cloud Score+	European Space Agency / Copernicus; Google	NDVI after the event vs normal years (crop loss)	Copernicus open licence. Pasquarella et al. (2023), Cloud Score+, CVPR Workshops
ESA WorldCover 2021 v200	European Space Agency	Cropland mask (class 40)	CC BY 4.0.
JRC Global Surface Water v1.4	European Commission JRC	Excludes rivers, lakes and reservoirs	CC BY 4.0. Pekel et al. (2016), Nature 540:418-422
Indian Village Boundaries, Maharashtra (Census 2001 villages)	DataMeet community	Village boundaries; talukas and districts are built by merging villages. Some taluka names and boundaries have changed since 2001.	ODbL 1.0.

Officials still verify and sign the panchanama; this list tells them where to go first.